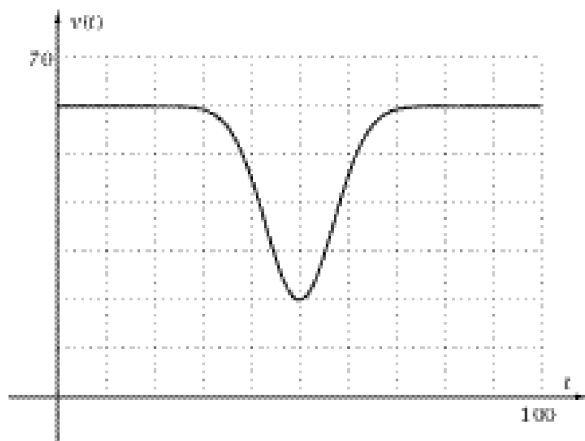


Quiz 1, Chapter 1

Objective: Show what you have learned about derivatives, and definite integrals so far.

Part 1: No calculators allowed (1–5)

For Problems 1–5, the velocity of a car, $v(t)$, in feet per second, is given as a function of t by this graph.



1. Draw a line tangent to the graph at the point on the graph where $t = 60$. Use it to estimate the instantaneous rate of change of $v(t)$ at that time. Which concept of calculus is this instantaneous rate?

2. How far did the car go between $t = 0$ and $t = 20$ s?

3. By counting squares on the graph, estimate the distance the car went between $t = 30$ and $t = 70$ s.

4. Draw four trapezoids on the given figure that you could use to estimate the distance in Problem 7.

5. Tell what might have happened in the real world to make the car's velocity-time graph look this way.

6. Derive the Trapezoid rule, starting with the formula for a trapezoid. Use an example exponential function with 4 trapezoids to assist in your derivation.

Flip the page and raise your hand to get your calculator for the next section)

Quiz 1, Chapter 1 continued

**Part 2: Graphing calculators allowed
(7–11)**

For Problems 7–11, Calvin and Phoebe start up a hill in their car. When they are 50 ft from the bottom of the hill, the car runs out of gas. They coast to a stop, then start rolling backward. Phoebe figures that their displacement, $D(t)$, in feet from the bottom of the hill, is given by

$$D(t) = -2t^2 + 20t + 50$$

7. How far are they from the bottom of the hill when $t = 3$ s?

8. Find Calvin and Phoebe's average velocity for the interval $t = 3$ to $t = 3.1$ s.

9. By using times closer and closer to $t = 3$, make a conjecture about the limit the average velocity approaches as t approaches 3.

10. The average velocity, $v_{av}(t)$, from 3 to t s can be written

$$v_{av}(t) = \frac{D(t) - D(3)}{t - 3}$$

Substitute appropriate expressions for $D(t)$ and $D(3)$. Then plot the graph of $v_{av}(t)$. Use a friendly window that includes $t = 3$. Sketch the graph, showing the behavior of the function at $t = 3$.

11. The instantaneous velocity at $t = 3$ would be the value of $v_{av}(t)$ at $t = 3$. Show what happens when you substitute $t = 3$, and thus tell why the instantaneous velocity cannot be found by direct substitution.